

Reagents

Collagenase Type 4 – Worthington (LS004186)

Soybean Trypsin Inhibitor – Roche (10109886001) - 100 µL aliquots of 1 mg/mL made up in DPBS.

DNAse - Applichem (A3778) - 10 µL aliquots in DPBS each with 250 U

DPBS (no Ca, no Mg) - ThermoFisher (14190144)

Bovine Serum Albumin - Sigma (A8806)

Collagenase Enzyme Mix (1 mL)

10 mg Coll. Type 4

100 µL of 1 mg/mL Soybean Trypsin Inhibitor (100 µg)

5 µL Dnase I (125 U/mL)

895µL DPBS

Equipment

gentleMACS dissociator (130-093-235) + gentleMACS C-tubes (130-093-237)

Centrifuge for 1.5 mL, 15 mL conicals

Pipettes and pipet tips

15, 50 ml Conicals (MLS)

1.5 mL tubes (MLS)

30 µM filters - Miltenyi (130-098-458)

Petri dishes (MLS)

Razor blades (MLS)

Ice bucket w/ice

Hemocytometers - InCyto Neubauer Improved (DHC-NO1-5)

Protocol

1. Mince tissue with sterile razor blade on petri dish on ice.
2. Add 25 mg minced testis tissue to 1 mL enzyme solution on ice in 1.5 mL tube.
3. While incubating on ice, triturate 10x every two minutes for 10 minutes total time, using p1000 pipet set to 700 µL.
4. After 10 mins transfer the solution to miltenyi C tube.
5. Run program brain_03 twice (in 4 °C cold room).
6. Quick spin C tube to ensure contents go to the bottom of the tube.
7. Continue trituating on ice in C-tube 10x every two minutes for 25 minutes total time.
8. After 25 total minutes add digest mix to 30 µm filter on sterile 15mL conical tube on ice and rinse filter with 3 mL 10% FBS/PBS.
9. Spin at 500g for 5 min at 4 °C.
10. Discard supernatant and resuspend in 10 mL ice-cold 0.04% BSA/PBS.
11. Spin at 500g for 5 min at 4 °C.
12. Discard supernatant and re-suspend in 200 µL ice-cold 0.04% BSA/PBS.
13. Adjust to concentration needed for 1,000 cells/µl for Chromium or 100 cells/µl for Drop-seq.